CONFAI 2026 · PLAKSHA UNIVERSITY · TRACK 06 — RESPONSIBLE & ETHICAL AI

Refuse to Hallucinate

A deterministic validity firewall for Indian official survey statistics

MoSPI's tools help you
find
the number. This firewall checks whether you are
allowed to say
it — and stops you, with a cited rule, when you are not.

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Poster + live demo · localhost:8001/demo · offline · no API key · no GPU · $0.00 / query

1 The problem

Retrieval over official PDFs is solved. **Governance is not.**

Unconstrained RAG over sovereign statistics fails in three ways — observed live in MoSPI's StatsDoc pilot (*anecdotal*, per `evaluation/statsdoc_audit/PROTOCOL.md`):

✗ **Definitional hallucination.** MPCE expanded as "Mean Per Capita Expenditure" instead of the statutory "Monthly Per Capita Consumption Expenditure."

✗ **Unguided cross-round arithmetic.** ~40% MPCE growth quoted across the HCES 2011-12 → 2023-24 break (paper → 3-visit CAPI) with no caveat.

✗ **Retrieval poisoning.** A poisoned document chunk can override guardrails and flip a verdict.

The thesis. You cannot fix a probabilistic system with more probabilistic prompts. The decision — *is this claim allowed?* — must come from **versioned, deterministic rules**, before generation. Models translate; rules judge.

WHERE IT SITS

A drop-in governance layer for journalists, fact-checkers, legislative researchers and statistical agencies — a standalone service/CLI **and** an MCP tool that other agents call as a governance oracle.

2 The firewall

1 **Question** — natural language, any register

↓

2 **Typed claim** — translator: models translate, never judge

↓

3 · Firewall — five stages, fixed order, first rule fires

- **Safety-A** — injection · EN / Hindi / Hinglish
- **Scope** — forecast · causal · politics
- **Comparability** — survey breaks
- **Metadata** — taxonomy check
- **Numeric** — verified rows only

↓

4 · SHA-256 freeze — computed **BEFORE** retrieval

```
hash = SHA-256(claim + decision + reason + rules)
```

↓

5 Retrieve (untrusted document text)

↓

6 **Templated render + citations** — WARN/REFUSE wording is fixed

Safety-B · freeze before retrieval + figure attestation

The verdict hash is locked before any document is read. A planted "ignore the caveat" chunk changes nothing — the hash is byte-identical with or without poisoning. A second hash then binds the **number you see** to the verdict and its source cell.

```
frozen_hash = SHA-256(claim ⊕ decision ⊕ sub_code ⊕ reason ⊕ rule_version)
```

```
attestation = SHA-256(frozen_hash ⊕ row_id ⊕ value ⊕ unit ⊕ pdf_sha256)
```

Field-guarded cache · no safe threshold

Danger-pair and paraphrase cosine ranges overlap on both measured embedders, so **no similarity threshold is safe**. Reuse only on exact critical-field match — a national row never answers a state/sex slice.

```
0 collisions across 19 danger pairs · 1708 verified rows (55 touched by the golden set)
```

3 Results

0.000 — hallucination rate — 0 of 36 must-abstain questions answered wrongly	**1.000** — abstention recall — every must-abstain question refused/warned
100/100 — verdict accuracy [Wilson 0.963–1.000]	**59**/59 — numeric exact match [0.939–1.000]
0/19 — danger-pair cache collisions	**1708**/1708 — verified rows with provenance
1.000 — Safety-A P/R/F1 · n=28 (EN · hi · hi-en)	**$0.00** — marginal cost · p50 < 200 ms, CPU-only

VIDHI VS STATSDOC — HONEST COMPARISON

CHECK	STATSDOC (ANECDOTAL)	VIDHI (MEASURED)
Definition	"Mean Per Capita…"	statutory "Monthly…"
Cross-round math	~40% growth, no caveat	halts with R_COMPARABILITY_HCES_METHOD
Poisoned context	overrides verdict	SHA-256 frozen pre-retrieval
Cost / latency	~$0.02–0.05 · 3.5–8 s	$0.00 · p50 < 200 ms

StatsDoc observations are anecdotal (live probing, no archived transcripts); Vidhi numbers are measured offline via `python vidhi/eval.py`. We do not claim a measured head-to-head victory.

HONEST LIMITS

- **Coverage is the limit, not logic** — 1708 verified rows: the full PLFS Highlights battery (11 survey rounds × usual + CWS × LFPR/WPR/UR × 15+/all-ages × 9 sector-sex cells of both committed annual reports), the TUS 2024 battery (9 ICATUS divisions × gender/sector/age × participation/per-participant/per-person, plus Tables 10–11 group rows), and HCES 2023-24 trend columns (2022-23 and 2011-12 legs). Uncovered slices (e.g. state-level labour figures, non-badge rounds) return `WARN_NO_DATA` + a coverage note, never a guess.
- Heuristic extraction breaks on exotic phrasing; an LLM translator port exists (`VIDHI_USE_LLM=1`), unmeasured.
- No load test yet; eSankhyiki live-API path inherits third-party cert-verification weakness (query path is offline).

Six guarantees — the soul of the system

1. Rules judge, models translate
2. Freeze before retrieval
3. No cosine threshold is safe
4. No PASS without provenance
5. Ambiguity is never contradiction
6. Honest coverage — never a guess

REPRODUCE EVERYTHING (OFFLINE)

```
# no API key · no GPU · no network
pip install -r requirements.txt
python -m pytest tests/ -q        # 110 passed · every store row reachable
python vidhi/eval.py               # 100 golden + 19 danger + 28 red-team + Safety-B
python -m uvicorn vidhi.service:app --port 8001  # → /demo
```

TRY THE LIVE DEMO

localhost:8001/demo — ask a PLFS / HCES / TUS / MIS question. Watch the verdict freeze, the citations resolve, and every figure name its source row.

MCP: `python -m vidhi.mcp_server` (4 governance tools) · **Agent:** `POST /analyze` for a policy briefing.

THE QUESTION THIS WORK WON'T HALLUCINATE

Ask the hard question — expect `WARN_NO_DATA` + a coverage note, never a guess. Inject in English, Hindi or Hinglish and watch Safety-A refuse with a cited rule.

Questions and red-team attempts welcome.